Fake News Detection System

System Architecture Specification & Technical Documentation — Category: Natural Language Processing & ML

Author: Rakshitha S (Full-Stack & AI/ML)

Institution: Jnana Vikas Inst of Tech, Bengaluru

Repository: https://github.com/rakshapatil526/Fake-News-Detection-System

Status: Verified Repository

Doc ID: SPEC-FAKE-NEWS-DETECTOR

Date: March 2026

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1. Domain Problem & Disinformation Analysis

The uncontrolled proliferation of digital disinformation and sensationalized fake news undermines public trust, skews elections, and promotes health hazards. The Fake News Detection System is an end-to-end NLP and machine learning platform that ingests news headlines and article bodies, extracts linguistic markers, and delivers real-time credibility classifications.

2. NLP Preprocessing & Feature Engineering Pipeline

• **Text Cleaning:** Conversion to lowercase, regex stripping of URLs, HTML tags, punctuation, and non-alphabetic characters.

• **Linguistic Normalization:** NLTK stop-word elimination and morphological root reduction via the Porter Stemmer.

• **Vectorization:** Scikit-learn `TfidfVectorizer` extracting unigram and bigram tokens with sublinear term-frequency scaling and minimum document frequency thresholds.

• **Linguistic Metrics:** Real-time calculation of lexical diversity, average sentence length, and clickbait keyword trigger frequencies.

3. Algorithm Selection & Dual Consensus Model

Algorithm	Mathematical Principle	Role in System
Multinomial Naive Bayes	Bayes Theorem with conditional word independence	Fast probabilistic baseline prioritizing vocabulary frequency
Logistic Regression	Sigmoid activation over linear hyperplane weights	Calibrated confidence score and robust feature weights
Dual Consensus Ensemble	Weighted vote of Naive Bayes + Logistic Regression	High-confidence detection minimizing false positive news flags

4. Interactive User Interface & Evaluation

• **Streamlit UI:** Real-time verdict badge (REAL vs. FAKE), confidence percentage meter, token preview, and Plotly confusion matrix.

• **Tech Stack:** Python 3.13, Streamlit 1.53, Scikit-learn, NLTK, Pandas, NumPy, Plotly, Joblib.